Sohan Jayram Reddy Kadiri

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PROFESSIONAL SUMMARY

Security Engineer with experience building vulnerability-management, cloud-security and secure distributed systems using Python, TypeScript and AWS. Skilled in vulnerability prioritization, IAM, application security, security automation and threat-informed remediation, with hands-on experience implementing least-privilege controls and secure-by-default architectures.

EDUCATION

Northeastern University Sep 2025 - May 2027
Master of Science, *Computer Science* GPA: 3.9

Siddaganga Institute of Technology Dec 2021 - Jun 2025
Bachelor of Technology, *Artificial Intelligence and Data Science*

PROJECTS

Vulnerability Management & Threat Prioritization Platform | Python, FastAPI, PostgreSQL, Redis, CVSS, EPSS, CISA KEV, SSVC, Docker

- Built a **centralized vulnerability-management platform** integrating findings from 6 infrastructure scanners and multiple SAST, SCA, DAST, container and IaC security sources.
- Engineered a **6-factor threat-prioritization model using CVSS, CISA KEV, EPSS, SSVC, exploit availability and asset criticality**, ranking 1,000+ vulnerabilities by remediation urgency.
- Automated **finding deduplication, threat enrichment and SLA-based remediation workflows**, reducing manual vulnerability-triage time by 60% across multi-scanner security data.

AWS Cloud Security Auto-Remediation | Python, AWS Config, EventBridge, Lambda, IAM, S3, CloudWatch, Terraform

- Built an **event-driven AWS security remediation pipeline** using AWS Config, EventBridge and Lambda to detect S3 public-access violations and automatically restore secure configuration.
- Enforced **least-privilege IAM and deterministic remediation**, restoring all 4 S3 Public Access Block controls after controlled non-compliance tests and returning affected resources to COMPLIANT state.
- Validated the complete **AWS Config → EventBridge → Lambda security path** through live AWS testing, CloudWatch audit logs and 18 automated tests with passing CI.

Payment Orchestration Platform — Secure Payment Microservices | TypeScript, PostgreSQL, RabbitMQ, Joi, Docker, GitHub Actions

- Implemented **fail-closed authentication and configuration controls** using API keys, internal service tokens and timing-safe comparisons, preventing services from starting with missing security-critical secrets.
- Designed **idempotency fencing, PostgreSQL row locks and ownership leases** to prevent duplicate, stale and conflicting payment operations during retries, crashes and concurrent requests.
- Enforced **secure-by-default API and service boundaries** through authentication-before-validation, restricted CORS, generated OpenAPI contracts and architecture tests integrated into CI.

WORK EXPERIENCE

Siemens | Software Engineer Intern | Remote Aug 2024 - Jul 2025

- Resolved **9 production defects through log analysis, API tracing and service-level debugging**, improving reliability across 2 applications serving 2,500+ users.
- Validated **authenticated REST API integrations and authorization boundaries**, tracing failed requests across frontend, backend and downstream services to isolate production defects.
- Led 3 engineering interns through **architecture review, API integration and production validation**, reducing recurring integration defects by 35% during an 8-week delivery cycle.

Unisys | Software Engineering Intern | Remote Apr 2024 - Jun 2024

- Engineered **real-time WebSocket messaging** supporting 350 concurrent connections with sub-150 ms latency and 99.8% message delivery.
- Diagnosed **API, database and client-side failures** across a marketplace serving 8,000+ users and processing 1,600+ orders, strengthening reliability of critical user workflows.
- Improved application resilience through **performance analysis and iterative optimization**, reducing page load time below 1.2 seconds and increasing mobile conversion by 8%.

Siddaganga Institute of Technology | Research Assistant | Tumkur May 2024 - Jan 2025

- Evaluated a **quantum-based 2Q learning approach for DDoS detection** across 5 datasets, achieving 95% detection performance.
- Improved model convergence by **30% through experimental optimization and comparative evaluation**, contributing to a peer-reviewed IEEE publication.

SKILLS

Languages: Python, Java, TypeScript, JavaScript, C++, SQL

Security Engineering: Vulnerability Management, Security Automation, Application Security, Cloud Security, IAM, RBAC, Authentication, Authorization, Secure API Design, Threat Modeling

Vulnerability & Threat Intelligence: CVSS, EPSS, CISA KEV, SSVC, CWE, Vulnerability Prioritization, Remediation Tracking

Security Testing: SAST, DAST, SCA, Container Scanning, IaC Scanning, Semgrep, CodeQL, OWASP ZAP, Trivy

Cloud & Infrastructure: AWS (IAM, Config, S3, Lambda, EventBridge, CloudWatch), Docker, Kubernetes, Terraform, Azure

Backend & DevSecOps: FastAPI, Node.js, PostgreSQL, Redis, RabbitMQ, REST APIs, GitHub Actions, CI/CD, Secrets Management, Logging

CERTIFICATIONS & ACHIEVEMENTS

- Open Source - Merged PR #472 into mindsdb/anton, fixing a bug that deleted saved user rules and adding 3 regression tests. Sep 2026
- Publication - "Detection of DDoS Attacks in Fog Computing using Interpretable Quantum-based 2Q Learning Approach" - IEEE Jan 2025